Shivendra Agrawal

Curriculum Vitae

1300 30th Street
Boulder, Colorado
(720) 421-5561
shivendra.agrawal@colorado.edu
https://www.linkedin.com

Summary - I am a second year **Ph.D student** at CU Boulder. Prior to this, I have experience at Noodle.ai and Bosch as a data scientist and as a co-founder of a start-up. I am broadly interested in Human Robotics Interaction, Assistive Technology, Computer Vision, 3D Scene Understanding, and Explainable AI. I am particularly interested in the application of these in Assistive Robotics.

Education

2019– **University of Colorado, Boulder**, *Ph.D. Cohort*, *Computer Science*.

- My work involves creating efficient, safe and explanable Human Robot Interaction (HRI) focused on helping visually impaired people.

2017–2019 **University of Colorado, Boulder**, *Masters of Science*, *Computer Science*.

- GPA: 4.0/4.0
- Teaching Assistant - Awarded **outstanding TA** for **Algorithms** (2017-2018, 2018-2019)
- Instructor - Awarded the **best part time instructor** 2019-2020 for the undergraduate Algorithms.

2009–2014 **Indian Institute of Technology, Kharagpur**, *Dual Degree*, *Mathematics Computing*.

Papers

September 2018 **Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning**, *Best paper nominee, HRI 2019*, CU Boulder.

We introduced a new framework to infer the human collaborators' mental view of the optimum state, predict future failure based on the inferred human mental model and provide interpretable feedbacks to help the human in a shared task. (Link to the paper)

Projects

Fall 2020 - **Smart Cane**, CU Boulder.

Currently working on building a smart cane that can operate in real world crowded public indoor settings. I am investigating if conveying longer plans are better or not. I am also interested in exploring ways to help create a better spatial mental model of the scene for the user who is visually impaired.

Fall 2020 - **Table top guidance for people who are visually impaired**, CU Boulder.

Currently working on creating optimization based algorithms for robotic manipulator to help a person with visual impairment work as a dyadic team on table top setting. I am primarily investigating the best modality for such a team to function based on various criteria like safety, speed, smoothness etc.

Feb 2019 - Apr 2019 **Robot guide dog for visually impaired**, Prof. Brad Hayes, CU Boulder.

In a first step towards my Ph.D. thesis we (in a team of 3) tried to emulate a guide dog in indoor scenario. We proposed a solution based on mobile robot platform. The robot would provide navigation assistance and scene explanation using a physical lease feedback and natural language interface. The goal is to help the visually impaired reach their goal faster and safer. (Video demo)

Feb 2019 - Apr 2019 **Visual memory with sparse demonstrations**, *Prof. Chris Heckman*, CU Boulder.

Explored learning a path information using visual cues extracted via a CNN so that the robot can re-trace the path without the 3D map of the environment and with very less demonstrations of the path.
Showed that a vanilla LSTM is still insufficient to generate crucial policies just from a few demonstration and may need attention going forward. (Link to project)

August 2018 - December 2018 **An ARKit app to help people with learning disabilities**, *Virtual Reality*, CU Boulder.
We made an IOS app with a python ML backend that parsed text from images and represented it in a more intuitive form that enabled learning. We also created a text editor which would help a person with dyslexia to just focus on some text.
My role - I designed the python backend, the IOS app, the text editor, came up with a way to translate from 2D text image coordinates to 3D world coordinates and helped conceptualize the product.(Link to project)

April 2018 – May 2018 **Logo Generation using GAN**, *Human Centered Machine Learning*, CU Boulder.
We (2 membered team) created a GAN based Logo generator to help generate original and unique initial logo designs. Tried out various variations like WGAN and ACGAN.
My role - Introduced a new way for labeling training data based on the color of the logo to help train ACGAN, developed the Web app that generates logos based on various criteria. (Link to report)

May 2018 **Twitter Trend Analysis**, *Big-Data Architecture*, CU Boulder.
We (5 membered team) developed a scalable web app using Kafka, Cassandra, and Spark which could classify upto 6k tweets per minute in real-time. Additionally, each trend was also classified using majority classification of it's tweets and similar trends were identified using Doc2Vec similarty metric.
My role - project conceptualization, trend similarity, leaflet and front-end development.(Video demo)

Jan 2018 **Grading Tool**, *Teaching Assistant*, CU Boulder.
Built Selenium based browser automation tool to automatically create prettified report cards from Google Sheets and post to nearly 300 students' Moodle or Canvas accounts.

Oct 2017 – Dec 2017 **Image Depth Map Estimation**, *Machine Learning*, CU Boulder.
Compared various CNN based architectures including transfer learning, residual layers and fully convolutional networks using Keras. Demonstrated an application of the predicted depth map to create *"Portrait Mode"*. (Link to report)

Oct 2017 – Dec 2017 **Motion Planning with RRT**, *Algorithms*, CU Boulder.
Created a Javascript based GUI implementation of RRT based algorithm for path finding. Compared this with vanilla A* varying multiple settings including sampling percentage, complexity of obstacles and tree expansion distance. (Link to report)

Nov 2015 – Feb 2016 **Splatter - Food recommender**, *Self*.
Co-created an android app based system for food recommendations and discovery based on preference, vicinity and friends. Designed the tag suggestion system, feed generation system (time-distance decay model for a dynamic feed) and notification system. (Link to App - No longer active)

Other professional experience

Reviewer.
HRI, ICRA, IROS, THRI

TA.
Algorithms, Human Computer Interactions, Intro to Robotics

Instructor.
Algorithms

Previous Work Experience

Sep 2016 – May 2017 **Associate Data Scientist**, Noodle.ai, Bangalore, India.

- ***Exploratory Data Analysis (EDA) tool*** - Developed a Django app to perform preliminary data analysis. The EDA tool automatically munges data to plot the features, performs statistical tests, find interesting inferences which may be overlooked if there are a lot of features/data and summarizes the result. The tool used parallelization whenever the operations don't need to be sequential.
- ***Demand forecast demo tool*** - Developed a demo tool which uses dummy data and shows micro-demand (store-level) predictions focusing FMCG companies. This tool was used a multiple times and proved to be more effective than a slideshow.
- ***Churn analysis*** - Created a model for churn prediction for a fashion e-commerce brand. Developed reusable, automated pipeline to perform all initial univariate/bivariate analysis and plotting.

Mar 2016 – Aug 2016 **Co-Founder**, PROGYRUS, Bangalore, India.

- Progyrus provided (Uber-like) large scaled human powered translations to help brands go vernacular. We worked with 6 clients to pilot regional content with 50 translators on the platform
- As one of the co-founders I contributed in conceptualizing the idea and launched the MVP product. Designed the order management and task management/allotment systems.
  Developed dashboards using AngularJS. Also used Django REST framework + PostgreSQL and developed APIs to automate the translation workflow and integration at client side.

June 2014 – Feb 2016 **Senior Engineer (Analytics)**, ROBERT BOSCH GMBH, Bangalore, India.

- **Prototype optimization for an automobile manufacturer** - Developed code to computationally form feasible prototypes (satisfying all boolean formulas) to reduce the number of actual test car productions. Devised and parallelized the algorithms that modeled the problem into a maximal clique finding problem.
- **Knowledge base for automobile domain** - Worked with a team to co-develop a system to gather and exploit unstructured knowledge from automobile specific publicly available domains/forums. The ontology aimed at assisting mechanic identify the symptom, problem and solution. Used dependency parsing, word2vec, fuzzy similarity and regular expressions to build the system.
- **Analysis for delay** - Developed a Python based GUI (tkinter) tool for the upper management to take data of purchase requests at regular interval and generate all insights into the delays and reveal all the bottlenecks and recurring issues.
- **Ticket Analysis** - Analyzed texts in complain tickets and came up with a novel hierarchical clustering method to identify common problems and sub-problems and made D3.js app.

Skills & Abilities

Languages	Python, Javascript, R, C++
Frameworks	AngularJS, Django REST, Flask
Libraries	Keras, Scikit-Learn, Selenium, Elasticsearch, PyTorch
Databases	MongoDB, PostgreSQL

Courses at CU Boulder

ML	ML, NLP, Human-Centered ML, Probabilistic Models
Robotics	Algorithmic Human Robot Interaction (AHRI)
Theory	Algorithms
Others	Virtual Reality & Augmented Reality, Big-Data Architecture

Interests

YouTube Channel	I have started a YouTube channel to explain some complicated algorithms and have had a good start with my first video. Checkout my channel at - (INFORMAL-CS)
Teaching	I enjoy teaching and I have taught study groups and under-privileged children apart from my teaching experiences at CU.
Food	I like cooking and exploring various cuisines and currently have 15 million total views of mainly food on Google Maps.
Other	I enjoy occasional biking and tennis.